YOU ZHANG

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EDUCATION

China Agricultural University Expected Jul 2027

Undergraduate in Computer Science and Technology | Beijing, China

Awards: First-Class Academic Scholarship, 2025 - 2026 (**3/68, Top 4.4%** in the academic year)

Second-Class Academic Scholarship, 2023 - 2024 (**6/101, Top 5.9%** in the academic year)

Selected Coursework: Operating Systems (A), Computer Architecture (A), Compiler Principles (A), Machine Learning (A), Data Structures (A), Artificial Intelligence (A)

RESEARCH INTERESTS

Efficient AI Systems · Inference Runtimes · Memory Management · Heterogeneous Computing

RESEARCH EXPERIENCE

MaizeGuard: Task-Aware Inference System Optimization **[GitHub]** Jul 2025 - Jun 2026

Undergraduate Innovation Project, China Agricultural University | Supervisor: Assoc. Prof. Qin Ma

- **Built a CPU/CUDA dual-ONNX system** for multi-ear maize anomaly analysis; developed an application-level runtime prototype with bounded in-flight tasks, CPU workers, and a device execution thread to overlap pipeline stages.
- **Applied stage-level profiling** to identify CPU geometric-processing and post-processing bottlenecks; implemented foreground-guided local morphology and component-local distance transforms and watershed splitting, preserving boundary semantics and verifying full-output equivalence.
- **Validated a 51.5% reduction in mean CPU pipeline latency** (1.46→0.71 s/image) for local processing versus a same-model full-frame reference, using 120 stratified native-resolution images and 20 paired passes on four physical Ryzen 9 7945HX cores.

  Project software registered in China (No. 2026SR0822662, 2026).

Class Definitions and Presentation Order in Zero-Shot LLM Classification May 2026 - Aug 2026

Independent Research

- **Designed controlled zero-shot text-classification experiments** with Phi-3-mini and Qwen2.5-1.5B to test how added class definitions and presentation order affect classification accuracy.
- **Separated development and confirmation samples;** used definition paraphrases and paired comparisons to assess wording sensitivity, with follow-up controls separating candidate-label order from definition-line order.
- **Identified an order-dependent reversal** on 7,600 confirmation examples from the AG News training split: adding definitions changed Phi-3-mini accuracy by -10.58 pp under one ordering and +7.39 pp under another, relative to prompts with class names and task instructions only. Completed a research manuscript.

RESEARCH EXPERIENCE

Multi-Agent Coordination with Reinforcement Learning and Local Planning Aug 2026

Individual Summer Research Project, Zhejiang University | Supervisor: Prof. Fei Miao

- **Trained and refined MAPPO** for cooperative navigation in MPE2 Simple Spread; used behavioral diagnosis and ablations to add training-time assignment guidance and continuous control, raising end-of-episode coverage success from 3.8% to 41.2%.
- **Reviewed collision avoidance and local planning methods** to refine communication-free local MPC; after tests showed excessive smoothing impaired arrival, added a bound on predicted arrival-cost increases.
- **Designed and compared three control schemes** - policies alone, baseline MPC, and MPC with an arrival-cost bound - to balance coverage and collision avoidance, using 1,500 simulation episodes with matched initial conditions. MPC with the bound raised coverage success from 50.6% to 61.6% and reduced collision rate from 3.176% to 2.117% versus policies alone.

WORK EXPERIENCE

360 Digital Security Group (Qihoo360)

AI Application Development Intern Jul 2026 - Aug 2026

Qwen-VL-Based Visual Data Pre-Annotation Pipeline

- Built a Qwen-VL object-detection pre-annotation pipeline integrating API calls, structured outputs, coordinate conversion, COCO annotation generation, and visual checks.
- Expanded the annotation schema from 3 to 14 classes and connected programmatic validation, platform import, and human review, supporting a single delivery of approximately 10,000 images.

AI Engineering Intern (Computer Vision & Edge Deployment) Jan 2026 - Mar 2026

YOLOv8 Object Detection and NCNN Edge Inference

- Built a 50-class object-detection training dataset for the team by sourcing, filtering, and processing 460,000+ images from Objects365 v1 and other datasets, including class mapping, COCO-to-YOLO conversion, and visual verification.
- Participated in multi-GPU YOLOv8n training using the prepared dataset; converted the trained model from PyTorch to FP32 NCNN and validated inference, completing the workflow from data preparation and training to NCNN inference.

SKILLS AND LANGUAGES

Programming and ML: Python, C/C++, SQL, PyTorch

Development Tools: Linux/Bash, Git, LaTeX

Languages: IELTS Academic 6.5; CET-6; CET-4